



When simplicity beats optimization: evidence from factor timing, volatility management, and the 1/N benchmark

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Abstract

This paper examines whether volatility management and factor-based mean–variance optimization improve out-of-sample portfolio performance relative to simple diversification benchmarks. Using nine equity factors from 1976 to 2025, the analysis compares recursive mean–variance portfolios, volatility-managed factors, equal-weight 1/N portfolios, and volatility-managed 1/N portfolios across multiple rolling windows, subperiods, and VIX regimes. Although volatility management improves the Sharpe ratios of several individual factors, these gains do not translate into robust portfolio-level outperformance once estimation risk and recursive implementation are considered. Optimized portfolios generally fail to be outperformed simple diversified benchmarks, while dynamic factor-selection strategies suffer from instability and look-ahead bias. Overall, the evidence suggests that simple diversification remains remarkably difficult to outperform in risk-adjusted terms, even when sophisticated volatility-management and optimization techniques are employed.

1 Introduction

In quantitative investment management, complexity is often treated as a synonym for sophistication. Factor models reduce the dimensionality of the investment universe, mean–variance optimization converts expected returns and covariances into portfolio weights, and volatility management adjusts exposures dynamically in response to changing risk. Each layer appears theoretically appealing. For hedge funds and institutional investors, however, the relevant question is not whether a strategy is elegant in sample, but whether it produces a higher and more stable Sharpe ratio out of sample. Cumulative returns can be mechanically scaled through leverage, but Sharpe ratios cannot be improved by leverage alone. This makes risk-adjusted per-

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formance the central metric for evaluating whether a quantitative strategy genuinely adds investment value.

The starting point of this paper is the long-standing tension between optimization and estimation risk. Markowitz's mean–variance optimization (MVO) framework provides the canonical foundation for portfolio choice by showing how diversification can improve the return–risk tradeoff (Markowitz 1952, 1959). Yet the same framework is highly sensitive to estimation error in expected returns and covariance matrices. Klein and Bawa (1976), Bawa et al. (1979), and Jorion (1985, 1988, 1992) show that parameter uncertainty can lead optimized portfolios to perform poorly out of sample, even when they appear optimal in sample. This problem is especially relevant for factor investing, where a relatively small set of factors may still generate unstable optimized weights when expected returns and covariances are estimated from finite samples.

Factor models offer one response to this problem by shrinking the investment universe from many assets to a smaller number of systematic return drivers. The empirical asset-pricing literature documents that characteristics such as size, value, profitability, investment, and momentum are associated with cross-sectional return patterns that are not fully explained by the market factor alone (Fama and French 1992, 1993, 2015; Hou et al. 2015, 2017). At the same time, the proliferation of factors creates its own challenge: a larger factor menu can increase opportunities for diversification, but it can also increase data mining, instability, and overfitting. Ang (2014) emphasizes the practical importance of disciplined factor investing, while Giglio et al. (2022) highlight how modern factor models and machine learning methods must confront the same fundamental problem of out-of-sample robustness.

Parametric portfolio policies provide a natural bridge between factor investing and portfolio optimization. Instead of optimizing over a large number of individual securities, Brandt et al. (2009) propose parameterizing portfolio weights as functions of characteristics or factor exposures. This approach is attractive because it keeps the economic structure of factor investing while reducing the dimensionality of the portfolio problem. In principle, such policies should help investors exploit cross-sectional factor premia while controlling estimation risk. In practice, however, the success of parametric policies depends on whether the estimated factor weights remain stable out of sample.

A second major development is volatility management. Moreira and Muir (2017) show that scaling exposures inversely with lagged volatility can improve Sharpe ratios for a range of factors. The intuition is simple: if expected returns do not rise proportionally during high-volatility periods, reducing exposure when volatility is high can improve risk-adjusted performance. DeMiguel et al. (2023) extend this question to a multifactor setting and show that the benefits of volatility-managed portfolios are more nuanced once realistic portfolio construction issues are considered. This raises an important practical question: does volatility management still add value when combined with parametric mean–variance portfolio policies, or does it simply introduce another layer of estimation error?

This paper studies that question by using nine widely used equity factors from 1976 to December 2025: Market, SMB, HML, RMW, CMA, UMD, ROE, IA, and BAB. The sample begins in 1976 to ensure common availability across the complete

factor set. The analysis compares original factors, volatility-managed factors, parametric mean–variance portfolios, equal-weight 1/N benchmarks, volatility-managed 1/N benchmarks, and smart volatility-management strategies. The empirical design is deliberately out-of-sample. Portfolio weights are estimated recursively using rolling windows, with robustness checks based on 36-, 60-, and 120-month estimation windows. The paper also evaluates performance across pre-2008, post-2009, and post-2020 subperiods, as well as across VIX regimes.

The results challenge the idea that more sophisticated factor timing automatically improves investment performance. The unconstrained parametric portfolio constructed from the nine original factors outperforms each individual factor in Sharpe ratio terms, indicating that factor combination is valuable relative to holding stand-alone factors. However, once the appropriate benchmark is introduced, the picture changes substantially. The simple 1/N portfolio is difficult to beat, and in many specifications, it outperforms mean–variance optimized parametric policies. This finding is consistent with the central message of DeMiguel et al. (2009): estimation error can make optimized portfolios less effective than naïve diversification.

The evidence on volatility management is also mixed. At the factor level, volatility management improves Sharpe ratios for some factors, but not uniformly across all nine factors. Volatility-managed 1/N portfolios provide a useful benchmark because they combine the simplicity of equal weighting with volatility scaling. Yet the main empirical conclusion is not that volatility management universally dominates. Rather, volatility management can improve risk-adjusted returns in selected settings, but its benefits are not robust enough to justify the claim that volatility-managed parametric mean–variance policies consistently dominate simple diversified benchmarks. In several robustness tests, 1/N or volatility-managed 1/N remains competitive or superior.

The paper further shows that estimation-window choice matters substantially. Shorter rolling windows such as 36 months often produce unstable mean–variance weights and weaker Sharpe ratios, while longer windows such as 120 months generate more stable results. This reinforces the interpretation that the central obstacle is not the absence of sophisticated portfolio tools, but the difficulty of estimating expected returns and covariances reliably. The results also show that smart selection rules, which choose volatility-managed factors only when they have recently outperformed original factors, are unstable in real time. Fixed full-sample selection appears more attractive, but this performance partly reflects look-ahead information and therefore cannot be treated as a genuine out-of-sample strategy.

The contribution of this paper is therefore threefold. First, it evaluates parametric factor portfolio policies against both standard 1/N and volatility-managed 1/N benchmarks, rather than only against individual factors. Second, it provides formal Sharpe-ratio comparison tests following the logic of Jobson and Korkie (1981) and Memmel (2003), addressing whether observed performance differences are statistically meaningful rather than merely point-estimate differences. Third, it shows that the interaction between volatility management and mean–variance optimization is highly sensitive to estimation windows and market regimes. The central conclusion is simple but important for investment practice: sophisticated factor timing and volatility scaling may improve performance in selected cases, but simple diversification remains a very difficult benchmark to beat.

The remainder of the paper is organized as follows. Section 2 reviews the literature on portfolio optimization, factor models, parametric portfolio policies, volatility management, and the 1/N benchmark. Section 3 describes the data, factor construction, summary statistics, and correlation structure. Section 4 presents the methodology, including the parametric portfolio policy, volatility-managed factors, smart volatility-management strategies, and the rolling out-of-sample design. Section 5 reports the empirical results, including Sharpe-ratio tests, robustness across rolling windows and subperiods, and VIX-regime analysis. Section 6 discusses the implications for hedge funds and institutional investors. Section 7 concludes.

2 Literature review and hypothesis development

2.1 Portfolio optimization, estimation risk, and the limits of mean–variance efficiency

The theoretical foundation of this study begins with mean–variance portfolio choice. Markowitz (1952, 1959) shows that investors can improve risk-adjusted performance by combining assets whose returns are imperfectly correlated. This framework remains the benchmark model for portfolio construction because it provides a direct link between expected returns, covariances, and optimal portfolio weights. In the mean–variance setting, the investor is not simply seeking high expected returns, but the best tradeoff between return and risk.

The practical difficulty is that mean–variance optimization requires estimates of expected returns and covariance matrices. These estimates are noisy, and small errors can lead to large changes in optimal weights. Klein and Bawa (1976) and Bawa et al. (1979) show that ignoring estimation risk can materially distort portfolio choice. Jorion (1985, 1988, 1992) further demonstrates that shrinkage and Bayesian methods can improve portfolio stability, but these approaches do not eliminate the fundamental problem that optimized portfolios are highly sensitive to parameter uncertainty. Black and Litterman (1990), Black (1992), He and Litterman (1999), and Litterman (2003) also respond to this instability by incorporating equilibrium-based priors and investor views into portfolio construction.

This literature implies an important tension. Optimization is theoretically attractive, but it can perform poorly out of sample when inputs are estimated imprecisely. This concern is central to the present paper because factor-based parametric policies are still mean–variance portfolios at their core. Even though the investment universe is reduced from individual securities to a small set of factors, the strategy must still estimate expected returns and covariances. Therefore, whether parametric factor portfolios outperform simple benchmarks is ultimately an empirical question.

2.2 Factor models and dimensionality reduction

Factor models provide a natural way to reduce the dimensionality of portfolio choice. Instead of modeling the returns of many individual securities directly, factor models describe asset returns through a limited number of common risk drivers. Ross

(1976) develops the arbitrage pricing theory as a flexible multifactor alternative to the CAPM. Huberman (1982), Chamberlain (1983), Chamberlain and Rothschild (1983), and Ingersoll (1984) further develop the theoretical basis for diversified factor structures in large asset markets. This line of work motivates the idea that systematic factors can summarize broad sources of return comovement.

Empirical asset pricing gives these factors concrete economic content. Fama and French (1992, 1993) show that size and value help explain stock returns beyond the market factor. Fama and French (2015) extend the model by adding profitability and investment factors. Hou et al. (2015, 2017) propose an investment-based approach that also emphasizes profitability and investment-related anomalies. Momentum and betting-against-beta further expand the practical factor menu used by asset managers. Ang (2014) presents factor investing as a systematic approach to asset management, while Cazalet and Roncalli (2014) and Clarke et al. (2016) discuss the practical implementation and efficiency of factor portfolios.

However, factor investing also creates new challenges. As the number of factors increases, the risk of overfitting and data mining rises. Giglio et al. (2022) emphasize that modern factor models and machine learning methods must be evaluated out of sample because apparent in-sample explanatory power may not translate into investable performance. McLean and Pontiff (2016) also show that return predictability can decline after publication, suggesting that factor premia may weaken once they become widely known. These findings are directly relevant to the present paper because a factor strategy that appears attractive over the full sample may not be robust across subperiods or rolling estimation windows.

2.3 The 1/N benchmark and Naïve diversification

The equal-weight 1/N portfolio is a central benchmark for evaluating optimized portfolio rules. DeMiguel et al. (2009) show that many optimized strategies fail to outperform 1/N out of sample. Their result is important because 1/N does not estimate expected returns or covariances. It is therefore immune to a large part of the estimation error that affects mean–variance portfolios. Although 1/N is simple, it can be difficult to beat precisely because it is robust.

The 1/N benchmark is especially relevant in a factor portfolio setting. If factors already represent diversified long–short portfolios, equal weighting across factors may provide substantial diversification without requiring noisy parameter estimates. Mean–variance optimization can improve performance only if estimated means and covariances contain enough reliable information to offset the additional estimation risk. This is not guaranteed. In fact, the more unstable the estimated covariance matrix or expected return vector becomes, the more likely it is that optimization underperforms simple diversification.

For hedge funds and institutional investors, this comparison is economically meaningful. A strategy that produces a higher cumulative return but a lower Sharpe ratio may simply be taking more risk. Since exposure can be levered, Sharpe ratio is the more informative measure of whether the strategy produces genuine risk-adjusted value. Therefore, the 1/N benchmark is not merely a naïve alternative; it is a demanding test of whether sophistication improves investable performance.

2.4 Parametric portfolio policies

Parametric portfolio policies offer a bridge between factor modeling and portfolio optimization. Brandt et al. (2009) argue that instead of estimating unrestricted asset weights, investors can specify portfolio weights as functions of observable characteristics. This approach reduces the number of parameters and allows portfolio choice to exploit economically meaningful structure. In a factor setting, the same logic applies: rather than optimizing over individual securities, the investor optimizes over a limited set of factor returns.

The appeal of parametric policies is that they may preserve the benefits of mean–variance optimization while reducing estimation error. If factor returns summarize broad sources of systematic risk, then combining factors through a parametric policy should generate a more efficient portfolio than holding any individual factor alone. This suggests that optimized factor portfolios may outperform standalone factors in Sharpe-ratio terms. At the same time, the estimation-risk literature warns that optimization can still be fragile. Even with only nine factors, expected returns and covariances may be unstable across samples, especially when rolling windows are short.

This leads to the first hypothesis:

Hypothesis 1 *Parametric mean–variance portfolios constructed from the nine original factors outperform individual factor portfolios in Sharpe-ratio terms, but they do not necessarily outperform simple equal-weight factor diversification.*

This hypothesis reflects the distinction between two benchmarks. Beating individual factors is a relatively weak test of value added because diversification alone can improve Sharpe ratios. Beating the equal-weight 1/N portfolio is a much stronger test because it asks whether estimation-based optimization adds value beyond simple diversification.

2.5 Volatility-managed portfolios

Volatility management introduces time-series conditioning into portfolio construction. Moreira and Muir (2017) show that scaling exposure inversely with lagged volatility can improve Sharpe ratios. The central intuition is that high volatility is not always compensated by proportionally higher expected returns. Reducing exposure during high-volatility states can therefore improve risk-adjusted performance.

This idea is directly relevant to factor investing because factor returns often experience large drawdowns during turbulent market periods. If volatility-managed factors reduce exposure during high-risk episodes, they may improve Sharpe ratios and downside behavior. DeMiguel et al. (2023) examine volatility-managed portfolios from a multifactor perspective and show that the benefits of volatility timing depend on the interaction among factors, estimation choices, and portfolio construction. This suggests that volatility management should not be evaluated only factor by factor. It must also be tested in the portfolio context.

Volatility management may improve performance through at least two channels. First, it can improve the return–volatility tradeoff of individual factors. Second, it

can change the correlation structure among factors. If volatility scaling reduces correlations during market stress, then volatility-managed factors may provide stronger diversification benefits. However, volatility management also introduces another layer of modeling. The strategy must estimate volatility, choose a scaling rule, and rebalance dynamically. These decisions can increase turnover, amplify sensitivity to sample periods, and create instability when combined with mean–variance optimization.

This motivates the second hypothesis:

Hypothesis 2 *Volatility management improves Sharpe ratios for some individual factors, but its benefits are heterogeneous and do not uniformly extend to all factors or all portfolio construction methods.*

This hypothesis allows volatility management to be useful without assuming that it dominates unconditionally. It is consistent with the possibility that volatility-managed factors perform well in some market environments but not others.

2.6 Combining volatility management with parametric portfolio policies

The key empirical question in this paper concerns the interaction between volatility management and parametric mean–variance policies. In principle, the combination is appealing. Volatility management uses time-series information to scale factor exposure, while parametric portfolio policies use cross-sectional information to allocate across factors. If both sources of information are reliable, their combination should improve Sharpe ratios.

In practice, the combination may be fragile. A volatility-managed factor is itself an estimated strategy because it depends on a volatility measure and scaling rule. A parametric portfolio built on volatility-managed factors then estimates means and covariances of these transformed returns. This creates a multi-layer estimation problem. The original factor portfolio requires estimates of factor means and covariances. The volatility-managed parametric portfolio additionally depends on volatility estimates and the interaction between scaled factors. As the estimation-risk literature suggests, each added layer of estimation can reduce out-of-sample robustness.

This paper therefore distinguishes among several strategies. The first is the original parametric policy based on the nine unmanaged factors. The second is a volatility-managed 1/N benchmark, which asks whether simple diversification combined with volatility scaling is sufficient. The third is ST1, which combines each original factor with its volatility-managed counterpart inside a rolling out-of-sample framework before constructing the parametric portfolio. The fourth is ST2, a smart selection rule that uses volatility-managed factors only when they outperform their original counterparts. This structure allows the paper to test whether performance comes from volatility management itself, from mean–variance optimization, or from simple diversification.

The third hypothesis follows:

Hypothesis 3 *More complex volatility-managed parametric policies do not consistently outperform simple 1/N and volatility-managed 1/N benchmarks after accounting for out-of-sample estimation risk.*

This hypothesis is deliberately stronger than asking whether volatility management improves individual factors. It evaluates whether the full strategy survives comparison with implementable benchmarks. It also reflects the central economic question: whether greater sophistication produces a higher Sharpe ratio for investors.

2.7 Rolling windows, market regimes, and robustness

Out-of-sample performance can depend heavily on the estimation window. DeMiguel et al. (2009) emphasize that optimized portfolio performance is sensitive to the amount of historical data used to estimate moments. Short windows may respond quickly to changing conditions but produce noisy estimates. Long windows may generate more stable estimates but adapt slowly to structural change. This tradeoff is especially important for mean–variance strategies because expected returns are difficult to estimate precisely.

Market regimes add another layer of complexity. McLean and Pontiff (2016) show that return predictability can weaken over time, implying that factor performance may differ across earlier and later periods. Volatility-managed strategies may also behave differently across volatility environments because they are specifically designed to reduce exposure when volatility is high. If volatility management works mainly during crises, then full-sample averages may hide important state dependence. VIX regimes therefore provide a useful way to evaluate whether performance differences are concentrated in low-, medium-, or high-volatility states.

Robustness analysis is essential for this paper because the central claims concern investable performance rather than in-sample fit. A strategy that works only for one rolling window or one subperiod is unlikely to be useful to hedge funds or institutional investors (see Ledoit and Wolf 2008). By comparing 36-, 60-, and 120-month rolling windows across full-sample, pre-2008, post-2009, and post-2020 periods, the paper evaluates whether the observed Sharpe-ratio differences are stable. The expectation from the estimation-risk literature is that shorter windows will make optimized portfolios less reliable, while simple benchmarks such as 1/N should be more stable.

2.8 Summary of the research gap

The existing literature provides strong foundations for each component of this paper. Mean–variance theory explains why optimization may improve diversification, while the estimation-risk literature explains why it may fail out of sample. Factor models reduce dimensionality, but factor proliferation raises concerns about instability and overfitting. Volatility management can improve Sharpe ratios, but its multifactor performance depends on implementation. The 1/N benchmark remains difficult to beat because it avoids parameter estimation.

The gap is that these ideas are rarely evaluated jointly in a single factor portfolio framework. Specifically, the literature does not fully answer whether volatility-

managed parametric factor portfolios outperform simple 1/N and volatility-managed 1/N benchmarks after formal Sharpe-ratio testing, rolling-window robustness checks, subperiod analysis, and volatility-regime evaluation. This paper fills that gap by comparing original factors, volatility-managed factors, parametric mean–variance policies, smart volatility-management rules, and simple diversification benchmarks in a unified out-of-sample setting.

3 Data

In this study, we choose nine factors that represent broad sources of systematic equity-return variation commonly used in empirical asset pricing and quantitative portfolio management. The Market factor captures overall equity market risk. SMB (small minus big) measures the size premium associated with small-cap stocks, while HML (high minus low) captures the value premium based on book-to-market ratios. RMW (robust minus weak) and CMA (conservative minus aggressive) represent profitability and investment effects from the Fama–French five-factor framework. UMD captures momentum, reflecting the tendency of past winners to continue outperforming past losers. ROE measures profitability using return-on-equity characteristics, IA captures investment-related anomalies, and BAB (betting against beta) reflects the low-beta anomaly documented in the factor-investing literature. Together, these nine factors provide exposure to multiple economically distinct dimensions of equity returns and therefore offer a suitable setting for evaluating diversification, volatility management, and parametric portfolio policies.

This study uses monthly factor return data from January 1976 to December 2025. The sample begins in 1976 because several non-Fama–French factors, particularly BAB, IA, and ROE, are not consistently available before that period. The factor universe contains nine widely used equity factors: Market, SMB, HML, RMW, CMA, UMD, ROE, IA, and BAB. The Market, SMB, HML, RMW, CMA, and UMD factors are obtained from Kenneth French’s data library, while BAB, IA, and ROE are collected from the AQR research database. Daily VIX data used for regime analysis are downloaded from the CBOE/FRED database and converted into monthly observations.

Table 1 reports summary statistics for the nine original factors. The Market factor exhibits the highest average return and volatility, while profitability- and momentum-related factors generally display higher Sharpe ratios. Figures 1 and 2 reports the correlation matrix of the original factors. Consistent with the factor-investing literature, the correlations are far from perfect (correlation -0.5 – $+0.5$), implying meaningful diversification opportunities. The correlation matrix of volatility-managed factors shows, on average, lower cross-factor correlations (closer to 0 than the correlation of original factors), suggesting that volatility management may improve diversification benefits by reducing co-movement during turbulent periods.

The empirical design is fully out-of-sample. Volatility-managed factors are constructed using a fixed 12-month rolling volatility estimate following the intuition of Moreira and Muir (2017). Portfolio weights are then estimated recursively

Table 1 Summary statistics of original factor returns

	Mean	Volatility	Sharpe ratio	Skewness	Kurtosis
Market	0.071	0.157	0.453	-0.498	1.624
SMB	0.015	0.107	0.144	0.503	4.905
HML	0.033	0.106	0.308	0.050	2.114
RMW	0.034	0.078	0.440	-0.230	10.142
CMA	0.032	0.073	0.431	0.294	1.423
UMD	0.071	0.147	0.480	-1.284	9.557
ROE	0.061	0.088	0.698	-0.882	5.507
IA	0.040	0.068	0.591	0.145	1.177
BAB	0.097	0.114	0.849	-0.451	4.035
1/N Portfolio	0.050	0.040	1.266	-0.258	6.387

This table reports annualized summary statistics for the nine original factors and the equal-weight 1/N factor portfolio. Mean and volatility are annualized from monthly returns. The Sharpe ratio is computed as annualized mean excess return divided by annualized volatility. Skewness and kurtosis are computed using monthly returns. The sample period is January 1976–December 2025

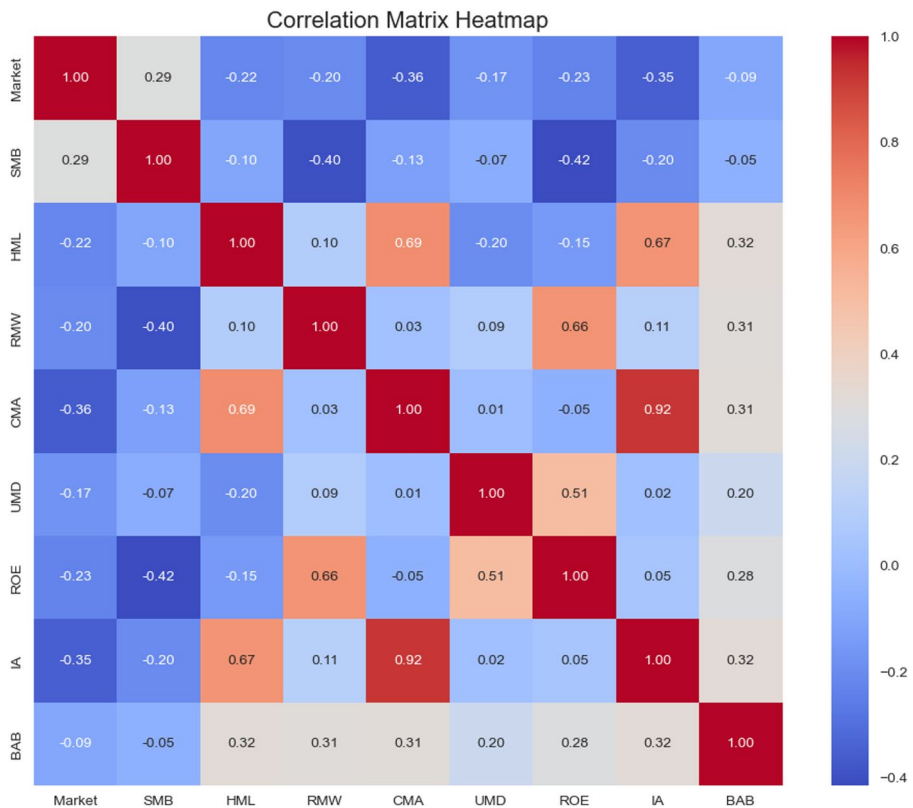


Fig. 1 Correlation matrix of original factor returns. *Note:* This figure reports the pairwise correlation matrix of the nine original factors. Warmer colors indicate higher positive correlations, while cooler colors indicate lower or negative correlations. The figure illustrates the degree of diversification available across the original factor universe

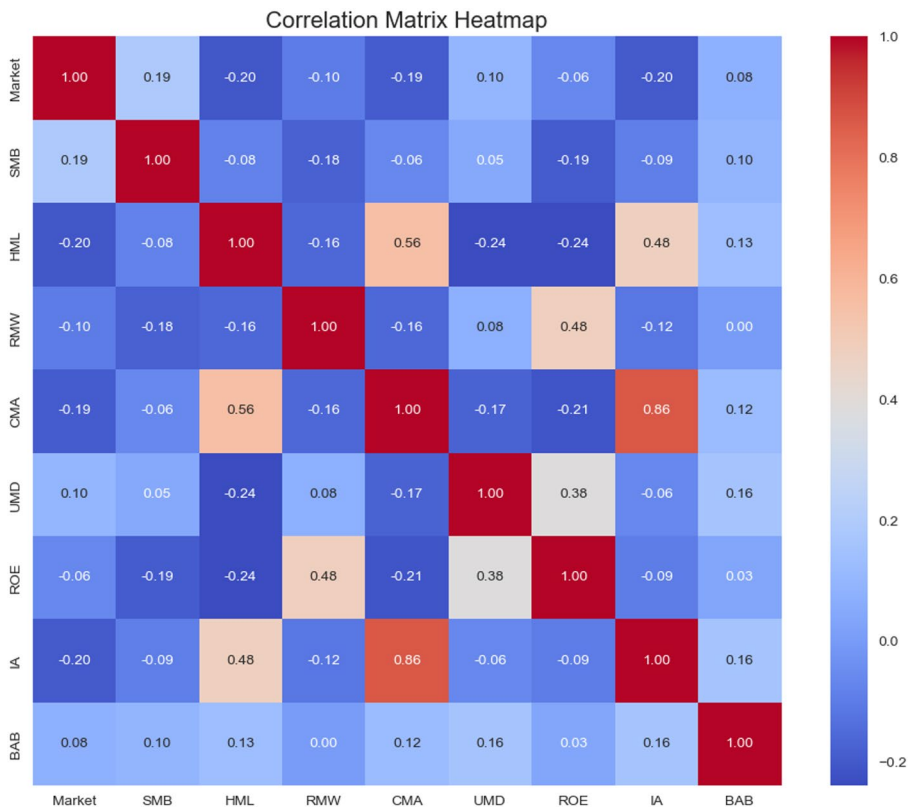


Fig. 2 Correlation matrix of volatility-managed factor returns. *Note:* This figure reports the pairwise correlation matrix of the volatility-managed versions of the nine factors. Volatility-managed factors are constructed using the approach stated in Sect. 4.3. Compared with Fig. 1, the matrix is used to evaluate whether volatility management reduces cross-factor co-movement and improves diversification potential

using rolling windows of 36, 60, and 120 months. Multiple rolling-window lengths are considered because the estimation-risk literature suggests a tradeoff between responsiveness and stability. Short windows react faster to changing market conditions but generate noisier parameter estimates, while longer windows provide more stable covariance structures but adapt more slowly to regime shifts. Due to differences in rolling-window requirements and data alignment across strategies, effective out-of-sample periods may vary slightly across comparisons. However, all pairwise Sharpe-ratio tests are conducted using overlapping observations only.

4 Methodology

4.1 Overview of the empirical design

The empirical framework evaluates whether volatility management and factor-based parametric portfolio policies can improve out-of-sample risk-adjusted performance relative to simple diversification benchmarks. The analysis proceeds in four stages.

First, volatility-managed (VM) versions of the nine factors are constructed using inverse-volatility scaling based on rolling 12-month volatility estimates. Second, parametric mean–variance portfolios are estimated recursively using rolling windows. Third, the study compares original factors, volatility-managed factors, equal-weight portfolios, and volatility-managed parametric portfolios using formal Sharpe-ratio tests. Finally, robustness is evaluated across alternative rolling-window lengths, subperiods, and VIX regimes.

The analysis focuses primarily on Sharpe ratios rather than cumulative returns. Since cumulative returns can be mechanically scaled through leverage, Sharpe ratios provide a more informative measure of risk-adjusted performance for hedge funds and institutional investors.

Note that in our research, the imported factors from Fama France website and AQR are excess returns already (no need to minus risk-free rate R_f again). Therefore, the mentioned return r all refers to excess return $\tilde{r}$ that minus R_f already in following content. It is worth mentioning that it is excess return rather than return that is commonly used for calculating Sharpe Ratio and conducting mean–variance optimization.

4.2 Factor returns and notation

Let $r_{i,t}$ denote the monthly return of factor i at time t , where

$$i \in \{MKT, SMB, HML, RMW, CMA, UMD, ROE, IA, BAB\}.$$

The factor return vector is

$$\mathbf{r}_t = (r_{1,t}, r_{2,t}, \dots, r_{9,t})'.$$

The sample contains monthly observations from January 1976 to December 2025.

4.3 Volatility-managed factors

Following the intuition of Moreira and Muir (2017), volatility-managed factors scale factor exposure inversely with lagged volatility.

For each factor i , rolling volatility is estimated using a fixed 12-month window:

$$\sigma_{i,t-1} = \sqrt{\frac{1}{12} \sum_{k=1}^{12} \left(r_{i,t-k} - \bar{r}_i \right)^2}.$$

The volatility-managed factor return is then

$$r_{i,t}^{VM} = \frac{c}{\sigma_{i,t-1}} r_{i,t},$$

where $r_{i,t}^{VM}$ is volatility-managed return of factor i ; $\sigma_{i,t-1}$ is rolling 12-month volatility estimate; c is scaling constant used to maintain comparable average exposure.

The purpose of volatility management is to reduce exposure during high-volatility periods and increase exposure during low-volatility periods. This may improve Sharpe ratios if expected returns do not rise proportionally with volatility.

4.4 Parametric portfolio policy

Parametric portfolio policy is used to link factor-level portfolio construction with the underlying cross-section of individual assets. Although the empirical implementation optimizes over nine factor returns, each factor itself represents a portfolio formed from a large number of underlying stocks. Therefore, the factor-level allocation can be interpreted as an indirect rule for assigning weights across the underlying N individual assets.

Let w_j denote the final portfolio weight on asset j , where $j = 1, \dots, N$. Suppose there are K factors, and each factor k implies a vector of asset-level weights:

$$\mathbf{w}_k = \begin{pmatrix} w_{k1} \\ w_{k2} \\ \vdots \\ w_{kN} \end{pmatrix}$$

The overall asset-level portfolio weight vector can then be written as a linear combination of factor-implied portfolios:

$$\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_N \end{pmatrix} = \theta_1 \begin{pmatrix} w_{11} \\ w_{12} \\ \vdots \\ w_{1N} \end{pmatrix} + \theta_2 \begin{pmatrix} w_{21} \\ w_{22} \\ \vdots \\ w_{2N} \end{pmatrix} + \dots + \theta_K \begin{pmatrix} w_{K1} \\ w_{K2} \\ \vdots \\ w_{KN} \end{pmatrix}$$

Equivalently,

$$\mathbf{w} = \sum_{k=1}^K \theta_k \mathbf{w}_k$$

where:

$\mathbf{w}$ = overall weight vector across N underlying assets;

$\mathbf{w}_k$ = asset – weight vector implied by factor k ;

θ_k = portfolio allocation to factor k .

This formulation shows why parametric policy is useful. Instead of estimating separate weights for thousands of individual stocks, the strategy estimates a small number of factor-level parameters, θ_k . These parameters determine how much capital is allocated to each factor portfolio, while the factor portfolios themselves determine how that exposure is distributed across the underlying assets.

In this study, the nine factor returns can therefore be interpreted as the returns on nine factor-implied portfolios. The parametric mean–variance policy estimates the optimal factor allocation vector:

$$\boldsymbol{\theta}_t = (\theta_{1,t}, \theta_{2,t}, \dots, \theta_{9,t})'$$

which is equivalent to the factor weight vector $\mathbf{w}_t$ used below. The resulting portfolio return is therefore generated by combining the nine factor portfolios rather than by directly optimizing over all individual stocks. This reduces dimensionality, improves interpretability, and connects the empirical strategy to economically meaningful sources of return.

4.5 Mean–variance optimized parametric portfolio policies

Portfolio weights are estimated recursively using rolling-window mean–variance optimization based on 9 factors.

At each time t , expected returns and covariance matrices are estimated using the previous M months of data:

$$\boldsymbol{\mu}_t = E_t(\mathbf{r}_t),$$

$$\boldsymbol{\Sigma}_t = \text{Cov}_t(\mathbf{r}_t),$$

where

$$M \in \{36, 60, 120\}.$$

In this study, 120 months are used for the main analysis, and 36/60 months are used in robustness checks.

The unconstrained portfolio weights are

$$\mathbf{w}_t = \frac{\boldsymbol{\Sigma}_t^{-1} \boldsymbol{\mu}_t}{\mathbf{1}' \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\mu}_t}.$$

This is the weights of a mean–variance optimized tangent portfolio, which maximizes the Sharpe ratio subject to the full-investment normalization:

$$\max_{\mathbf{w}_t} \frac{\mathbf{w}_t' \boldsymbol{\mu}_t}{\sqrt{\mathbf{w}_t' \boldsymbol{\Sigma}_t \mathbf{w}_t}}$$

subject to

$$1' \mathbf{w}_t = 1$$

The out-of-sample portfolio return is

$$r_{t+1}^P = \mathbf{w}_t' \mathbf{r}_{t+1}.$$

In addition to the unconstrained specification, the study also considers a constrained portfolio policy. The constrained specification imposes a market-exposure normalization. Specifically, the market factor weight is fixed at one:

$$w_{MKT,t} = 1$$

while the remaining factor weights are allowed to vary freely (obtained by doing mean–variance optimization):

$$\mathbf{w}_t^C = (1, w_{SMB,t}, w_{HML,t}, w_{RMW,t}, w_{CMA,t}, w_{UMD,t}, w_{ROE,t}, w_{IA,t}, w_{BAB,t})'$$

The constrained portfolio return is therefore:

$$r_{t+1}^C = r_{MKT,t+1} + \sum_{i \neq MKT} w_{i,t}^C r_{i,t+1}$$

This constraint is used because it makes the strategy easier to interpret as a market-anchored factor overlay. Instead of allowing the optimizer to scale the entire portfolio arbitrarily or take unstable market exposure, the market component is fixed at one, and the parametric policy only determines the additional tilts toward the other factors. This provides a cleaner comparison between the market benchmark and factor-based enhancements.

However, the empirical results show that this constrained policy produces weaker out-of-sample Sharpe ratios than the unconstrained tangency portfolio. The main results therefore focus on the unconstrained specification, while the constrained version is treated as a robustness check.

4.6 Equal-weight benchmarks

Two benchmark portfolios are constructed.

The standard equal-weight portfolio is

$$r_t^{EW} = \frac{1}{N} \sum_{i=1}^N r_{i,t}.$$

The volatility-managed equal-weight portfolio is

$$r_t^{VMEW} = \frac{1}{N} \sum_{i=1}^N r_{i,t}^{VM}.$$

where in this research $N = 9$.

These benchmarks are important because they avoid parameter estimation and therefore provide a direct test of whether optimization genuinely adds value beyond simple diversification.

4.7 ST1: rolling combination of original and volatility-managed factors

The first volatility-management strategy (ST1) combines each original factor with its volatility-managed counterpart inside a rolling out-of-sample framework.

For each factor i , a two-asset mean–variance optimization is estimated recursively:

$$\mathbf{x}_{i,t} = (r_{i,t}, r_{i,t}^{VM})'.$$

The rolling combination weights are

$$\mathbf{a}_{i,t} = \frac{\Sigma_{x,i,t}^{-1} \boldsymbol{\mu}_{x,i,t}}{1' \Sigma_{x,i,t}^{-1} \boldsymbol{\mu}_{x,i,t}}.$$

The combined factor return is

$$r_{i,t+1}^{ST1} = \mathbf{a}_{i,t}' \mathbf{x}_{i,t+1}.$$

The resulting combined factors are then used to construct the parametric portfolio described in Sect. 4.4.

Importantly, all ST1 weights are estimated recursively within rolling windows. This avoids look-ahead bias because future information is not used in factor construction.

4.8 ST2: Smart volatility-management selection

The second strategy (ST2) selectively replaces original factors with volatility-managed factors when volatility management improves Sharpe ratios.

Two implementations are considered.

The first (main ST2) is a rolling dynamic selection strategy, where the choice between original and volatility-managed factors is updated recursively using rolling windows. The second (ST2_fix) is a fixed full-sample selection strategy that selects volatility-managed factors using full-sample Sharpe-ratio comparisons and keeps those selections fixed through time.

Rolling Sharpe ratios are estimated as:

$$SR_{i,t}^O = \frac{E_t(r_{i,t})}{\sigma_t(r_{i,t})}$$

and

$$SR_{i,t}^{VM} = \frac{E_t(r_{i,t}^{VM})}{\sigma_t(r_{i,t}^{VM})}.$$

The selected factor is defined as:

$$r_{i,t+1}^{ST2} = \begin{cases} r_{i,t+1}^{VM}, & \text{if } SR_{i,t}^{VM} > SR_{i,t}^O, \\ r_{i,t+1}, & \text{otherwise.} \end{cases}$$

The rolling implementation represents the true out-of-sample strategy, while the fixed implementation contain look-ahead information and is therefore interpreted cautiously. Therefore, in following sections, ST2 refers to the rolling dynamic selection strategy.

4.9 Sharpe-ratio evaluation

Strategy performance is evaluated primarily using annualized Sharpe ratios (the ratio of expected excess return to volatility of excess return:

$$SR = \frac{E(r_t)}{\sigma(r_t)} \times \sqrt{12}.$$

To formally test whether two strategies have statistically different Sharpe ratios, the Jobson and Korkie (1981) test with the Memmel correction is applied. For two return series $r_{1,t}$ and $r_{2,t}$, the annualized Sharpe ratios are:

$$SR_1 = \frac{r_1}{\sigma_1} \sqrt{12}$$

$$SR_2 = \frac{r_2}{\sigma_2} \sqrt{12}$$

The null hypothesis is:

$$H_0 : SR_1 - SR_2 = 0$$

The corrected test statistic is:

$$Z_{JKM} = \frac{SR_1 - SR_2}{\sqrt{\frac{1}{T} [2(1 - \rho_{12}) + \frac{1}{2} (SR_1^2 + SR_2^2 - 2SR_1SR_2\rho_{12}^2)]}}$$

where T is the number of overlapping return observations, and ρ_{12} is the correlation between the two strategy return series. A statistically significant value of Z_{JKM} indicates that the difference in Sharpe ratios is unlikely to be driven by sampling variation.

The analysis evaluates whether:

- Market-weight constrained portfolios outperform unconstrained counterparties
- Parametric portfolios outperform individual factors;
- Optimized portfolios outperform 1/N benchmarks;
- Volatility management improves risk-adjusted performance;
- More sophisticated strategies outperform simple diversification.

4.10 Robustness analysis

Robustness is evaluated along three dimensions.

First, rolling-window lengths of 36, 60, and 120 months are compared. Short windows may respond more quickly to changing market conditions but generate noisier parameter estimates, while longer windows may produce more stable covariance estimates.

Second, the sample is divided into:

- Pre-2008
- Post-2009
- Post-2020

This allows the analysis to evaluate whether results differ across major market environments, including the Global Financial Crisis and the post-COVID regime.

Third, VIX regime analysis is conducted using terciles of monthly VIX levels:

- Low-VIX regime
- Medium-VIX regime
- High-VIX regime

Because rolling estimation windows differ across strategies and subperiods, effective out-of-sample sample lengths may vary slightly across comparisons. This does not affect the main conclusions because all pairwise Sharpe-ratio comparisons use overlapping observations only. The robustness analysis therefore strengthens the reliability of the results by evaluating whether the findings remain stable across alternative estimation horizons and market environments.

5 Empirical results and analysis

All data-processing scripts, portfolio-construction code, robustness tests, and replication files are available at: [GitHub link: https://github.com/XUAN-FENG9/Quant-Finance-and-Investment-Research/tree/main/FMPM_underperformance%20of%20vo]

l-managed%20PPP]. The empirical results in this paper are fully replicable using the data and code provided in the repository.

5.1 Unconstrained versus constrained mean–variance optimization

Table 2 compares the out-of-sample performance of unconstrained and constrained mean–variance optimization (MVO) portfolios. The constrained specification performs 90% substantially worse and generates economically unstable results. Although the market-factor exposure is fixed at one, the remaining factor weights are still estimated recursively using rolling covariance matrices. When the covariance matrix is noisy, the residual factor allocations become unstable and may produce extreme overlay positions.

By contrast, the unconstrained specification behaves more consistently and delivers higher out-of-sample Sharpe ratios. This result is economically intuitive because the unconstrained portfolio corresponds to the rolling tangency portfolio estimated from the factor return vector. The constrained portfolio introduces an additional restriction without materially reducing estimation risk.

Given the weak performance of the constrained specification, the remaining analysis focuses primarily on the unconstrained MVO framework.

5.2 Volatility management at the individual factor level

Table 3 compares the original and volatility-managed (VM) versions of the nine factors. The evidence is mixed. Volatility management 99% significantly improves the Sharpe ratios of several factors, particularly RMW, UMD, and ROE, and BAB.

These results suggest that volatility management does not uniformly improve factor performance. Instead, the effectiveness of volatility scaling appears factor-specific. For some factors, reducing exposure during high-volatility periods improves downside protection sufficiently to increase risk-adjusted returns. For other factors, volatility scaling may reduce exposure during periods when factor premia remain strong, thereby weakening performance.

An additional finding emerges from the factor correlation analysis. From Fig. 2 in Sect. 3, the average pairwise correlation among volatility-managed factors is lower than among the original factors. This indicates that volatility management may improve diversification benefits by reducing factor co-movement during turbulent

Table 2 Out-of-sample performance of unconstrained and constrained MVO portfolios

Unconstrained sharpe ratio	Constrained sharpe ratio	SR difference (uncon-cons)	<i>P</i> -value
0.986	0.872	0.114*	0.0664

This table compares the out-of-sample Sharpe ratios of unconstrained and constrained mean–variance optimization portfolios. The constrained specification fixes the market factor exposure at one, while the remaining factor weights are estimated recursively. “SR Difference” is computed as the unconstrained Sharpe ratio minus the constrained Sharpe ratio. The *P*-value is based on the Jobson–Korkie test with Memmel correction. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Effective sample lengths may differ slightly from other tables because rolling-window requirements determine the available out-of-sample observations

Table 3 Sharpe ratio comparison between original and volatility-managed factors

	Original sharpe ratio	VM sharpe ratio	SR difference (VM–Original)	P-value
Market	0.453	0.382	(0.071)**	0.025
SMB	0.144	0.075	(0.069)	0.480
HML	0.308	0.129	(0.178)***	0.000
RMW	0.440	0.467	0.027	0.500
CMA	0.431	0.308	(0.124)***	0.000
UMD	0.480	0.683	0.202***	0.000
ROE	0.698	0.953	0.255***	0.000
IA	0.591	0.512	(0.079)***	0.001
BAB	0.849	0.843	(0.006)	0.732

This table compares each original factor with its volatility-managed counterpart. “SR Difference” is computed as the volatility-managed factor Sharpe ratio minus the original factor Sharpe ratio. *Parentheses indicate negative differences.* P-values are obtained from the Jobson–Korkie test with Memmel correction. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The comparison uses the overlapping sample available after constructing volatility-managed factors

Table 4 Performance of 1/N and volatility-managed 1/N portfolios

VM 1/N sharpe ratio	1/N sharpe ratio	SR difference	Z-statistic	P-value
1.320595	1.202	0.118**	2.297	0.022

This table compares the standard equal-weight 1/N portfolio with the volatility-managed 1/N portfolio. Both portfolios allocate equally across the nine factor portfolios, while VM 1/N uses volatility-managed factor returns. “SR Difference” is computed as the VM 1/N Sharpe ratio minus the standard 1/N Sharpe ratio. The P-value is based on the Jobson–Korkie test with Memmel correction. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively

market conditions. However, lower correlations alone are not sufficient to guarantee superior portfolio performance once estimation risk and recursive optimization are considered.

5.3 Does volatility management improve diversified portfolios?

Table 4 evaluates the equal-weight benchmark portfolios. Over the full sample, the VM 1/N portfolio achieves a Sharpe ratio of 1.32, compared with 1.20 for the standard 1/N portfolio. The Jobson–Korkie test indicates statistical significance at the 95% level, but not at the 99% level.

Economically, this suggests that volatility management may provide some benefit when combined with simple diversification. However, the statistical evidence is not sufficiently strong to conclude that the improvement is uniformly robust. The relatively weak significance level (95%) implies that part of the observed difference may reflect sampling variation.

More importantly, the equal-weight results establish a demanding benchmark for the remaining analysis. Since both 1/N and VM 1/N avoid recursive factor covariance-matrix inversion, they are naturally less exposed to estimation error than rolling MVO portfolios. Consequently, the key question is not whether MVO can outperform individual factors, but whether it can outperform simple diversified benchmarks.

5.4 Smart volatility-management strategies and look-ahead bias

Table 5 compares the main portfolio strategies: the original MVO portfolio, ST1, ST2 rolling, ST2 fixed, 1/N, and VM 1/N.

A particularly important result is the sharp difference between ST2 (rolling) and ST2 fixed. ST2 fixed substantially far outperforms ST2 rolling because it selects volatility-managed factors using full-sample Sharpe ratios. In contrast, ST2 rolling updates the selection rule recursively using only past information.

This distinction is economically important. The strong performance of ST2 fixed is likely driven mostly by look-ahead bias because future information influences the factor-selection rule. Therefore, ST2 fixed should not be interpreted as a feasible real-time trading strategy.

Table 6 reports the factor-selection frequency under the rolling ST2 rule. The selection frequency is calculated as:

$$Freq_i = \frac{1}{T} \sum_{t=1}^T 1(i \in S_t)$$

where: $Freq_i$ is the selection frequency of factor i ; T is the total number of out-of-sample rolling selection dates; S_t is the set of factors selected for volatility management at time t ; $1(i \in S_t)$ equals one if factor i is selected as a VM factor at time t , and zero otherwise.

The most frequently selected VM factors are ROE, UMD, BAB, and RMW. Interestingly, these are exactly the same four factors selected by the fixed full-sample ST2

Table 5 Main strategy comparison: original MVO, ST1, ST2, 1/N, and VM 1/N

Panel A: Sharpe Ratios of different strategies						
	Original MVO	ST1	ST2 rolling	ST2 fix	1/N	VM 1/N
Sharpe Ratio	0.96	0.99	0.55	1.26	1.20	1.32
Panel B: Difference in Sharpe Ratios between two strategies (row-column) and the significance						
Strategy	Original MVO	ST1	ST2 Rolling	ST2 fix	1/N	VM 1/N
Original MVO		0.03	0.41***	(0.30)***	(0.24)***	(0.36)***
ST1			0.44***	(0.27)***	(0.21)***	(0.33)***
ST2 Rolling				(0.71)***	(0.65)***	(0.77)***
ST2 fix					0.06	(0.06)
1/N						(0.12)**

Panel A reports out-of-sample Sharpe ratios for the main strategies. Original MVO is the unconstrained rolling mean–variance portfolio based on the nine original factors. ST1 combines each original factor with its volatility-managed counterpart using rolling optimization. ST2 Rolling selects between original and volatility-managed factors recursively using only past information. ST2 Fixed selects volatility-managed factors using full-sample Sharpe ratios and is therefore reported only as an ex-post comparison subject to look-ahead bias. Panel B reports pairwise Sharpe-ratio differences, calculated as row strategy minus column strategy. *Parentheses indicate negative differences.* *P*-values are based on the Jobson–Korkie test with Memmel correction. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Because different strategies require different rolling windows and factor-construction periods, Sharpe ratios may differ slightly from those reported in other tables when the overlapping evaluation sample changes

Table 6 Selection frequency of volatility-managed factors in rolling ST2

Factor	Select frequency
ROE	0.899
UMD	0.751
BAB	0.438
RMW	0.386
HML	0.292
Market	0.277
IA	0.212
SMB	0.209
CMA	0.203

This table reports how frequently each factor is selected as a volatility-managed factor under the rolling ST2 rule. Selection frequency is calculated as the number of rolling dates on which a factor is selected divided by the total number of rolling selection dates. Higher values indicate that the volatility-managed version of the factor more frequently outperforms the original version within the rolling estimation window

specification. This indicates that the rolling strategy consistently identifies similar factors over time. However, despite selecting similar factors, the rolling implementation performs substantially worse.

One possible explanation is that dynamic switching between original and VM factors introduces additional estimation error and timing instability. Even if volatility management improves the long-run Sharpe ratio of some factors, recursively changing the factor specification may generate noisy allocation changes that weaken out-of-sample performance.

Table 5 further shows that ST1 (Sharpe ratio 0.99) performs much better than ST2 rolling (0.56). This suggests that continuously combining original and VM factors through rolling optimization is more stable than using a discrete switching rule. However, ST1 still fails to outperform the simple diversified benchmarks.

5.5 Main portfolio comparison: Does VM MVO outperform?

The main empirical results provide little evidence that volatility-managed MVO strategies outperform simple diversification.

Table 5 shows that ST1 achieves a Sharpe ratio of approximately 0.99, which is lower than both VM 1/N at 1.28 and standard 1/N at 1.19. The Jobson–Korkie tests indicate that ST1 99% significantly underperforms both benchmark portfolios.

Similarly, the original unconstrained MVO portfolio performs poorly relative to the diversified benchmarks. Although the original MVO portfolio outperforms most individual factors, it generally fails to outperform either 1/N or VM 1/N once estimation risk is taken into account.

The comparison between ST1 and the original MVO portfolio is also economically weak. ST1 slightly improves the Sharpe ratio relative to the original MVO portfolio, but the difference is statistically insignificant. Therefore, introducing volatility management inside the MVO framework does not materially improve risk-adjusted performance.

Overall, the main empirical evidence suggests that simple diversification is more robust than complex recursive optimization strategies in this setting. Volatility management may improve selected individual factors and may modestly improve equal-weight diversification, but these benefits are not strong enough to produce consistent outperformance once combined with rolling mean–variance optimization.

6 Robustness analysis

6.1 Robustness across subperiods and rolling windows

Table 7 reports robustness results across alternative subperiods and rolling estimation windows. The analysis considers 36-, 60-, and 120-month rolling windows within the full sample, pre-2008, post-2009, and post-2020 periods.

The results show that rolling-window choice is economically important. Shorter windows, particularly the 36-month specification, generally produce unstable Sharpe ratios for both the original MVO portfolio and ST1. This is consistent with the estimation-risk literature: shorter windows react more quickly to changing market conditions but generate noisier covariance estimates and less stable portfolio weights.

Longer rolling windows, especially the 120-month specification, produce more stable results. In some post-2009 specifications, MVO performance becomes more competitive relative to the diversified benchmarks. However, these cases are concentrated in relatively short effective out-of-sample periods because long rolling windows remove many early observations.

A particularly important finding is that VM 1/N does not consistently outperform the standard 1/N benchmark. In many specifications, the Sharpe ratios of the two portfolios are extremely close, and the Jobson–Korkie tests fail to reject equality even at the 95% level. In several recent-period specifications (such as post-2009 sub-period using 120 M window), VM 1/N even underperforms standard 1/N.

Table 7 also shows that ST1 rarely outperforms the original MVO portfolio in a statistically meaningful way. This reinforces the earlier conclusion that volatility management does not reliably improve recursive MVO performance once estimation risk is considered.

Figure 3 presents Sharpe-ratio comparisons across rolling windows and subperiods. The figure visually confirms that simple diversification strategies are substantially more stable across different estimation horizons, while MVO-based strategies remain highly sensitive to both rolling-window choice and market regime.

Post-2020 result is confusing and less interpretable as after cutting the beginning training window, the Sharpe ratios of strategies are based on only 35 data points, as shown in Table 7.

6.2 Robustness across VIX regimes

Table 8 reports the VIX-regime analysis. The sample is divided into low-, medium-, and high-VIX states using terciles of monthly VIX observations.

Table 7 Robustness across subperiods and rolling estimation windows

Panel A: Full Sample						
Window	Strategy	Sharpe ratio	Difference vs. 1/N	Difference vs. VM 1/N	Difference vs. original	Observations
120 M	Original	0.99	(0.22)***	(0.33)	–	586
120 M	ST1	0.99	(0.20)***	(0.29)	0.03	574
60 M	Original	0.00	(1.23)***	(1.29)	–	646
60 M	ST1	0.21	(1.02)***	(1.08)	0.21***	634
36 M	Original	0.35	(0.89)***	(0.92)	–	670
36 M	ST1	0.12	(1.11)***	(1.15)	(0.23)***	658
Panel B: Pre-2008 subperiod						
Window	Strategy	Sharpe ratio	Difference vs. 1/N	Difference vs. VM 1/N	Difference vs. original	Observations
120 M	Original	1.49	(0.04)	(0.06)	–	370
120 M	ST1	1.39	(0.12)	(0.10)	(0.05)	358
60 M	Original	1.18	(0.35)***	(0.29)	–	430
60 M	ST1	0.11	(1.42)***	(1.36)	(1.06)***	418
36 M	Original	0.35	(1.18)***	(1.10)	–	454
36 M	ST1	0.16	(1.37)***	(1.29)	(0.18)***	442
Panel C: Post-2009 subperiod						
Window	Strategy	Sharpe ratio	Difference vs. 1/N	Difference vs. VM 1/N	Difference vs. original	Observations
120 M	Original	0.56	0.07	0.44	–	83
120 M	ST1	0.25	(0.23)**	0.13	(0.30)***	83
60 M	Original	(0.27)	(0.82)***	(0.92)	–	143
60 M	ST1	0.39	(0.16)	(0.26)	(0.66)***	143
36 M	Original	0.50	(0.24)*	(0.42)	–	167
36 M	ST1	0.28	(0.46)***	(0.64)	(0.22)**	167
Panel D: Post-2020 subperiod						
Window	Strategy	Sharpe ratio	Difference vs. 1/N	Difference vs. VM 1/N	Difference vs. original	Observations
36 M	Original	0.74	1.09***	0.96	–	35
36 M	ST1	(0.26)	0.10	(0.03)	(0.99)***	35
Panel E: 1/N vs VM 1/N						
Subperiod	Window	1/N Sharpe ratio	Difference Vs. VM 1/N (1/N - VM 1/N)			Observations
full	120 M	1.20	(0.12)**			586
full	60 M	1.23	(0.06)			646
full	36 M	1.24	(0.04)			670
Pre-2008	120 M	1.52	(0.02)			370
Pre-2008	60 M	1.53	0.06			430
Pre-2008	36 M	1.53	0.09			454
Post-2009	120 M	0.49	0.37***			83
Post-2009	60 M	0.55	(0.10)			143

Table 7 (continued)

Panel E: 1/N vs VM 1/N

Subperiod	Window	1/N Sharpe ratio	Difference Vs. VM 1/N (1/N - VM 1/N)	Observations
Post-2009	36 M	0.74	(0.17)**	167
Post-2020	36 M	(0.36)	(0.14)*	35

This table reports out-of-sample Sharpe ratios and Sharpe-ratio differences across subperiods and rolling-window lengths. The subperiods are Full Sample, Pre-2008, Post-2009, and Post-2020. Rolling estimation windows are 36, 60, and 120 months where sufficient observations are available. “Difference vs 1/N,” “Difference vs VM 1/N,” and “Difference vs Original” report Sharpe-ratio differences relative to the corresponding benchmark. *Parentheses indicate negative differences.* *P*-values are based on the Jobson–Korkie test with Memmel correction. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Effective observations vary across panels because rolling estimation windows and subperiod definitions reduce the available out-of-sample sample. All pairwise tests are based on overlapping observations only

The results provide little evidence that portfolio management becomes substantially more difficult during high-volatility periods. VM 1/N does not significantly outperform the standard 1/N benchmark in any VIX regime. In the high-VIX regime, VM 1/N even produces a negative Sharpe ratio, while the standard 1/N portfolio maintains a positive Sharpe ratio.

This result is economically important because volatility management is often motivated by the idea that reducing exposure during turbulent periods should improve risk-adjusted performance. The empirical evidence here suggests that this intuition does not translate into robust out-of-sample gains in the current multifactor setting.

The original MVO portfolio also underperforms the diversified benchmarks across most volatility regimes. ST1 performs even worse in medium- and high-VIX states, indicating that recursively combining volatility-managed factors with MVO may amplify estimation instability during volatile periods.

Figure 4 visually confirms these patterns. Across all VIX regimes, simple diversification remains more stable than recursive optimization strategies. Overall, the robustness evidence strongly supports the main conclusion of the paper: sophisticated volatility-managed MVO strategies do not consistently outperform simple diversified benchmarks once estimation risk, rolling-window instability, and market-regime variation are taken into account.

7 Discussion and economic insights

The empirical results of this paper provide a consistent and economically important message: simple diversification is extremely difficult to outperform once estimation risk and realistic out-of-sample implementation are taken into account. Although volatility management and parametric portfolio policies are theoretically appealing, their practical benefits are much weaker than expected in the current multifactor setting.

A central finding is that the equal-weight benchmark remains remarkably competitive. Both the standard 1/N portfolio and the volatility-managed 1/N portfolio consistently outperform or match the more sophisticated recursive mean–variance optimization (MVO) strategies. This result is closely related to the estimation-risk literature. Mean–variance optimization requires estimating expected returns and cova-



Fig. 3 Sharpe ratios across subperiods and rolling estimation windows. *Note:* This figure plots Sharpe ratios for Original MVO, ST1, 1/N, and VM 1/N across different subperiods and rolling-window lengths. Each panel corresponds to one subperiod-window combination. Missing panels occur when the subperiod is too short to support the required rolling-window estimation. The figure highlights the sensitivity of recursive MVO strategies to estimation-window length and market regime

riance matrices, both of which are highly unstable in finite samples. Even when the number of assets is reduced from thousands of stocks to nine factors, the optimization problem remains sensitive to noise in the estimated moments. As a result, the theoretically optimal portfolio can become economically unstable out of sample.

The results therefore reinforce the argument of DeMiguel et al. (2009): the practical difficulty of portfolio optimization is not the optimization procedure itself, but the quality and stability of the estimated inputs. The current paper extends this insight into a multifactor volatility-management setting. The evidence suggests that adding more layers of estimation does not necessarily improve performance. Instead, additional modeling complexity may amplify instability.

This interpretation is particularly visible in the comparison between ST2 fixed and ST2 rolling. The fixed strategy performs substantially better because it uses full-sam-

Table 8 Strategy performance across VIX regimes

VIX regime	Strategy	Sharpe ratio	Difference vs. 1/N	Difference vs. VM 1/N	Difference vs. Original	Observations
Low	Original	1.61	(0.25)*	(0.23)	–	144
Low	ST1	1.66	(0.19)	(0.17)	0.05	144
Low	1/N	1.86	–	0.02	0.25*	144
Low	VM 1/N	1.84	(0.02)	–	0.23	144
Medium	Original	1.38	(0.13)	(0.14)	–	143
Medium	ST1	0.87	(0.64)***	(0.65)***	(0.51)***	143
Medium	1/N	1.51		(0.01)	0.13	143
Medium	VM 1/N	1.52	0.01	–	0.14	143
High	Original	0.23	(0.21)**	0.26***	–	144
High	ST1	0.03	(0.41)***	0.05	(0.20)***	144
High	1/N	0.44		0.46***	0.21**	144
High	VM 1/N	(0.03)	(0.46)***	–	(0.26)***	144

This table reports Sharpe ratios across low-, medium-, and high-VIX regimes. VIX regimes are defined using terciles of monthly VIX observations. The table compares Original MVO, ST1, 1/N, and VM 1/N. “Difference vs 1/N,” “Difference vs VM 1/N,” and “Difference vs Original” report Sharpe-ratio differences relative to each benchmark. *Parentheses indicate negative differences.* *P*-values are based on the Jobson–Korkie test with Memmel correction. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. All comparisons use overlapping observations within each VIX regime



Fig. 4 Sharpe ratios across VIX regimes. *Note:* This figure plots Sharpe ratios for Original MVO, ST1, 1/N, and VM 1/N across low-, medium-, and high-VIX regimes. The figure illustrates whether volatility management and recursive MVO strategies perform differently across market-volatility states

ple information to determine which factors should be volatility-managed. Once the strategy is implemented recursively in real time, performance deteriorates sharply. Economically, this result is important because it highlights the difference between ex-post optimization and feasible investment implementation. Many sophisticated strategies may appear successful when evaluated with hindsight, but much of the apparent alpha disappears once future information is removed.

The results also provide insight into the role of volatility management itself. At the individual factor level, volatility management improves the Sharpe ratios of several factors, particularly BAB, RMW, UMD, and ROE. In addition, the correlation structure among volatility-managed factors becomes weaker on average, suggesting that volatility scaling may improve diversification benefits during volatile periods. However, these improvements are not sufficiently strong or stable to generate consistent outperformance once volatility-managed factors are embedded inside recursive MVO portfolios.

This distinction is economically meaningful. Volatility management may help individual factors by reducing exposure during turbulent periods, but recursive MVO introduces another layer of estimation error through repeated covariance-matrix inversion. The interaction between volatility management and MVO therefore becomes fragile. In practice, the incremental benefit of volatility scaling is often offset by the instability of rolling portfolio optimization.

The robustness analysis further strengthens this interpretation. Rolling-window choice has a substantial effect on performance. Short estimation windows, especially the 36-month specification, produce highly unstable results. This occurs because short windows respond quickly to changing market conditions but generate noisy covariance estimates and rapidly changing portfolio weights. Longer windows, particularly the 120-month specification, produce more stable Sharpe ratios and occasionally allow MVO strategies to become more competitive relative to the diversified benchmarks.

This result has an important economic implication. The apparent value of optimization depends critically on the tradeoff between adaptability and estimation stability. In unstable markets, reacting too aggressively to recent observations may increase noise rather than improve forecasts. Long-horizon estimation windows may partially reduce this problem because they stabilize covariance estimates, although they also adapt more slowly to structural change.

The VIX-regime analysis provides additional evidence against the idea that volatility management universally improves portfolio performance. One might expect volatility-managed portfolios to perform particularly well during high-volatility periods. However, the empirical results do not support this expectation. VM 1/N fails to significantly outperform the standard 1/N benchmark in any VIX regime and even underperforms in the high-VIX state.

This finding suggests that high volatility alone does not guarantee that reducing exposure will improve risk-adjusted returns. During some market crises, factor premia may remain sufficiently strong that volatility scaling reduces returns more than risk. In other cases, volatility spikes may reverse quickly, causing volatility-managed portfolios to reduce exposure precisely when expected returns subsequently recover. Therefore, the economic effectiveness of volatility management depends not only

on volatility itself, but also on the interaction between volatility persistence, factor premia, and market timing.

Another important implication concerns the role of complexity in quantitative investment management. Modern asset management increasingly relies on sophisticated statistical models, machine learning methods, and dynamic allocation rules. However, the current evidence suggests that complexity should not automatically be interpreted as investment superiority. In many cases, simple diversified portfolios provide more stable and more robust risk-adjusted performance than highly optimized recursive strategies.

This does not imply that optimization or volatility management are useless. Instead, the results indicate that their incremental value may be smaller than often assumed once realistic implementation frictions and estimation uncertainty are considered. Volatility management may still improve selected factors, and optimization may still be useful in specific market environments or with better forecasting information. However, the evidence strongly suggests that simple diversification remains an extremely difficult benchmark to beat.

For hedge funds and institutional investors, the practical implication is clear. A sophisticated strategy should not be evaluated only by its theoretical elegance or in-sample fit. The key criterion is whether the strategy produces stable out-of-sample Sharpe-ratio improvements after accounting for estimation risk, recursive implementation, and changing market conditions. In the current empirical setting, the answer is generally negative. The diversified 1/N benchmark, despite its simplicity, remains more robust than most complex volatility-managed MVO strategies.

Overall, the paper highlights a broader lesson in quantitative finance: the main challenge is often not constructing sophisticated models, but ensuring that those models remain reliable when implemented in real time under uncertainty.

8 Conclusion

This paper examines whether volatility management and factor-based parametric mean–variance optimization (MVO) improve out-of-sample portfolio performance relative to simple diversification benchmarks. Using nine widely used equity factors from 1976 to 2025, the analysis compares original factor portfolios, volatility-managed factors, equal-weight benchmarks, and recursive rolling MVO strategies across alternative rolling windows, subperiods, and VIX regimes.

The empirical evidence provides little support for the view that more sophisticated portfolio construction methods consistently outperform simple diversification. Although volatility management improves the Sharpe ratios of several individual factors and slightly improves the volatility-managed equal-weight portfolio in some specifications, these gains are not sufficiently robust once recursive MVO estimation and out-of-sample implementation are considered. In most cases, the standard 1/N portfolio and the volatility-managed 1/N portfolio outperform or match the more complex MVO-based strategies.

The results also show that recursive optimization is highly sensitive to estimation-window choice. Short rolling windows generate unstable covariance estimates and

weak out-of-sample performance, while longer windows produce more stable but still limited improvements. Similarly, smart volatility-management selection strategies perform well only when future information is implicitly used, indicating that a substantial portion of the apparent alpha disappears once look-ahead bias is removed.

Overall, the findings reinforce the broader conclusion that estimation risk remains one of the central practical limitations of portfolio optimization. Even in a reduced-dimensionality factor setting, sophisticated recursive allocation rules struggle to generate stable Sharpe-ratio improvements relative to simple diversified benchmarks.

Several limitations should be acknowledged. First, the analysis focuses on U.S. equity factors and does not consider international assets or alternative asset classes. Second, transaction costs, turnover constraints, and leverage constraints are not explicitly modeled. Third, expected returns are estimated using historical rolling means rather than predictive macroeconomic or machine-learning signals.

Future research may extend the analysis by incorporating transaction costs, Bayesian or shrinkage estimators, machine-learning-based expected return forecasts, and multi-asset allocation settings. Another promising direction would be to examine whether volatility management becomes more effective when combined with regime-switching models or alternative risk measures beyond variance alone.

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Author contributions Dr. Xuan FENG is mainly in charge of methodology design, data collection & main analysis, manuscript written, revision, and supervision. Xuan Feng, PhD, CFA, FRM, FDP, PMP, is an independent researcher focusing on empirical asset pricing, portfolio construction, investment strategies, asset allocation, hedge funds, and financial risk management. His research is closely connected to practical investment problems, with particular emphasis on return predictability, risk-controlled portfolio design, strategy evaluation, and the implementation of systematic investment approaches. Dr. Feng is also a financial engineering practitioner with expertise in fixed income, quantitative pricing, and interest-rate derivatives. He has developed a famous publicly available GitHub tutorial repository [<https://github.com/XUAN-FENG9/Rates-Pricing-Quant-Library>] on interest-rate model pricing, which serves as a detailed learning hub for interest-rate models and related derivative pricing frameworks. This work reflects his broader interest in linking rigorous quantitative finance theory with practical model implementation. Dr. Feng earned his PhD in Finance from EDHEC Business School and subsequently worked as a quantitative model risk expert at a leading investment bank. The research presented in this paper was conducted independently. For more information, please see the author's LinkedIn profile: [<https://www.linkedin.com/in/dr-xuan-feng-217556a4/>].

Data availability This section outlines the data and factors utilized in our research, spanning from February 1967 to December 2025. The data is sourced from the public Fama–French website, providing a comprehensive dataset for our empirical analysis.

Declarations

Conflict of interest The authors declare no competing interests.

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